

MAX ZHU

Atlanta, GA | max.zhu@gatech.edu | 404-940-3568 | www.linkedin.com/in/maxzzhu/

EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S. in Computer Engineering and Mathematics, Minor in Applications of AI/ML

Expected Graduation: May 2027

- **Concentrations:** Systems & Architecture, Computing Hardware & Emerging Architecture **GPA: 3.91/4.00**
- **Relevant Coursework:** Data Structures and Algorithms, Object Oriented Programming, Intro to Software Engineering, Operating Systems, Digital System Design, HW/SW Programming, Processor Design, VLSI Design, Embedded Systems

SKILLS

Languages: Java, C++, C, Python, SQL, SystemVerilog, RISC-V, Perl, MATLAB, HTML

Frameworks: Django, React, Jupyter Notebook, Node, Kubernetes, MongoDB, MySQL, Git, Docker

EXPERIENCE

FUSMobile

Apr 2025 – Aug 2025

Embedded and QA Engineer

Alpharetta, GA

- Engineered a bidirectional USB serial protocol between an Arduino Due and an imaging workstation to enable closed-loop automatic self-alignment, parsing live X-ray inputs from the Neurolyzer to drive real-time spatial orientation
- Built an automated medical report generation system in C++, using MFC for metadata extraction, OpenCV for image filtering/processing, and libHaru for annotation, reducing manual reporting time by 90%
- Re-engineered the Neurolyzer, increasing portability by 20% and interactivity, for hands-on conference demonstrations

Intro to Software Engineering

Jan 2025 – May 2025

Scrum Master

Atlanta, GA

- Led a team of 5 in an Agile development cycle as Full-Stack Lead, while adapting deliverables to evolving client requirements
- Engineered a DIY display for Spotify using an ESP8266, C++, and the Spotify Developer API, for real-time playback
- Implemented OAuth 2.0 to fetch live playback data and personalized user profiles
- Developed an interactive web-app, deployed on Heroku, mimicking Spotify Wrapped, providing personalized profile analytics, AI-driven listening summaries, and animated visuals
- Created an Atlanta Food Finder leveraging Django + React + Google Maps API to help users explore local cuisine, bookmark favorite restaurants, and receive tailored dining recommendations

MARGA Lab

Aug 2024 – May 2025

Wireless Digital Twin Researcher

Atlanta, GA

- Created Python scripts with Sionna to optimize transmitter-receiver placement by over 60% in over 50 unique environments
- Analyzed and annotated 30+ research papers on 5G, Wi-Fi 6, ray tracing, and UAVs to establish baseline models
- Published and presented a paper on varying geometric simplifications applied to Blender environments for WDTs, reducing simulation rendering time by 50% while preserving over 95% signal fidelity compared to full-scale models

Big Data Big Impact

Sept 2024 – August 2025

Software Engineer

Atlanta, GA

- Conducted time-series modelling and anomaly detection for watershed and water-affordability datasets, using Jupyter Notebooks and Keras classifier models to identify areas of risk in communities
- Extracted, cleaned, and transformed public datasets for ML pipelines, generating insights for UN Clean Water goals
- Leveraged AI to build a RAG pipeline with integrated sentiment analysis and stock-chart visualization, transforming traditional earnings call transcripts into an interactive conversational platform for financial insight

Silicon Jackets

Aug 2024 – Present

Digital Design and Verification Engineer

Atlanta, GA

- Driving RTL development and verification across two tapeouts for a 100MHz RV32I ALU, shifter, and L1 data-cache pipeline
- Authoring micro-architectural specifications and block diagrams, translating system requirements into synthesizable SystemVerilog
- Established a robust verification infrastructure using a UVM env (agents/scoreboard) and an ISS reference to achieve 100% functional and $\geq 95\%$ code coverage, revealing 10 edge cases through automated Python/CI regression pipelines

EXTRACURRICULARS

Leadership: GT IEEE (Director of Corporate Affairs), Hive (Master Peer Instructor), ECE Student Ambassadors

Clubs: Silicon Jackets (Digital Designer), Chinese Student Association (Dynasty Head), Trading@GT (Prediction Markets), Phi Delta Theta

Interests: Motorcycles, Weightlifting, Algorithmic Trading, Cars, Snowboarding, Keyboards, Photography